

# MODULE 1

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## 1.1 SCALABLE COMPUTING OVER THE INTERNET

Over the past 60 years, computing technology has undergone a series of platform and environment changes. In this section, we assess evolutionary changes in machine architecture, operating system platform, network connectivity, and application workload. Instead of using a centralized computer to solve computational problems, a parallel and distributed computing system uses multiple computers to solve large-scale problems over the Internet. Thus, distributed computing becomes data-intensive and network-centric. This section identifies the applications of modern computer systems that practice parallel and distributed computing. These large-scale Internet applications have significantly enhanced the quality of life and information services in society today.

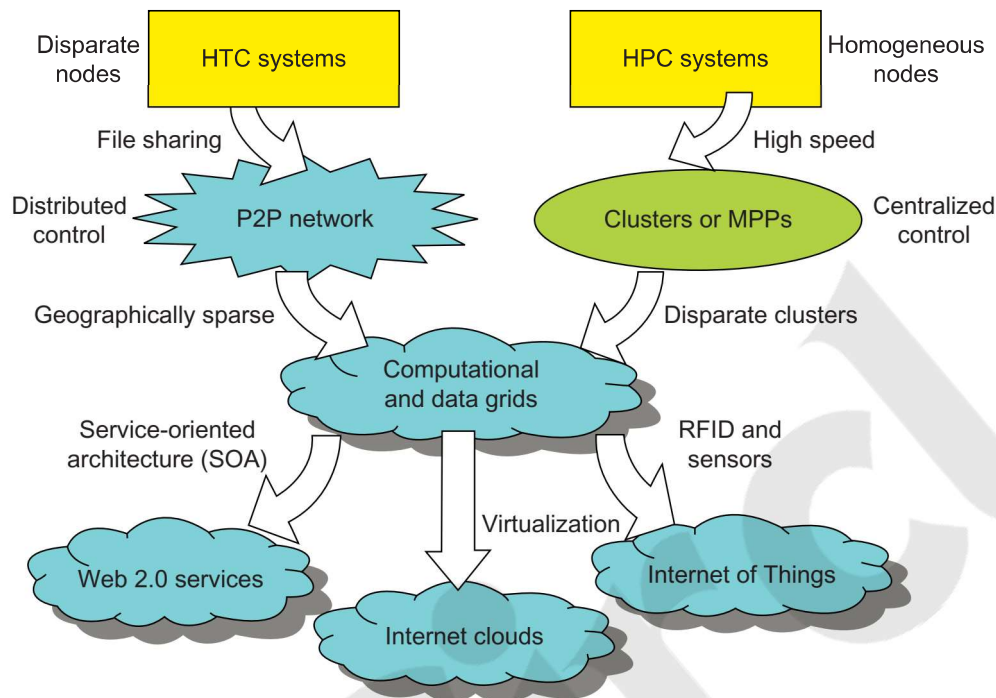
### 1.1.1 The Age of Internet Computing

Billions of people use the Internet every day. As a result, supercomputer sites and large data centers must provide high-performance computing services to huge numbers of Internet users concurrently. Because of this high demand, the Linpack Benchmark for *high-performance computing (HPC)* applications is no longer optimal for measuring system performance. The emergence of computing clouds instead demands *high-throughput computing (HTC)* systems built with parallel and distributed computing technologies [5,6,19,25]. We have to upgrade data centers using fast servers, storage systems, and high-bandwidth networks. The purpose is to advance network-based computing and web services with the emerging new technologies.

#### 1.1.1.1 The Platform Evolution

Computer technology has gone through five generations of development, with each generation lasting from 10 to 20 years. Successive generations are overlapped in about 10 years. For instance, from 1950 to 1970, a handful of mainframes, including the IBM 360 and CDC 6400, were built to satisfy the demands of large businesses and government organizations. From 1960 to 1980, lower-cost mini-computers such as the DEC PDP 11 and VAX Series became popular among small businesses and on college campuses.

From 1970 to 1990, we saw widespread use of personal computers built with VLSI microprocessors. From 1980 to 2000, massive numbers of portable computers and pervasive devices appeared in both wired and wireless applications. Since 1990, the use of both HPC and HTC systems hidden in

**FIGURE 1.1**

Evolutionary trend toward parallel, distributed, and cloud computing with clusters, MPPs, P2P networks, grids, clouds, web services, and the Internet of Things.

clusters, grids, or Internet clouds has proliferated. These systems are employed by both consumers and high-end web-scale computing and information services.

The general computing trend is to leverage shared web resources and massive amounts of data over the Internet. Figure 1.1 illustrates the evolution of HPC and HTC systems. On the HPC side, supercomputers (*massively parallel processors* or *MPPs*) are gradually replaced by clusters of cooperative computers out of a desire to share computing resources. The cluster is often a collection of homogeneous compute nodes that are physically connected in close range to one another. We will discuss clusters, MPPs, and grid systems in more detail in Chapters 2 and 7.

On the HTC side, *peer-to-peer (P2P)* networks are formed for distributed file sharing and content delivery applications. A P2P system is built over many client machines (a concept we will discuss further in Chapter 5). Peer machines are globally distributed in nature. P2P, cloud computing, and web service platforms are more focused on HTC applications than on HPC applications. Clustering and P2P technologies lead to the development of computational grids or data grids.

### 1.1.1.2 High-Performance Computing

For many years, HPC systems emphasize the raw speed performance. The speed of HPC systems has increased from Gflops in the early 1990s to now Pflops in 2010. This improvement was driven mainly by the demands from scientific, engineering, and manufacturing communities. For example,

the Top 500 most powerful computer systems in the world are measured by floating-point speed in Linpack benchmark results. However, the number of supercomputer users is limited to less than 10% of all computer users. Today, the majority of computer users are using desktop computers or large servers when they conduct Internet searches and market-driven computing tasks.

### 1.1.1.3 High-Throughput Computing

The development of market-oriented high-end computing systems is undergoing a strategic change from an HPC paradigm to an HTC paradigm. This HTC paradigm pays more attention to high-flux computing. The main application for high-flux computing is in Internet searches and web services by millions or more users simultaneously. The performance goal thus shifts to measure *high throughput* or the number of tasks completed per unit of time. HTC technology needs to not only improve in terms of batch processing speed, but also address the acute problems of cost, energy savings, security, and reliability at many data and enterprise computing centers. This book will address both HPC and HTC systems to meet the demands of all computer users.

### 1.1.1.4 Three New Computing Paradigms

As [Figure 1.1](#) illustrates, with the introduction of SOA, Web 2.0 services become available. Advances in virtualization make it possible to see the growth of Internet clouds as a new computing paradigm. The maturity of *radio-frequency identification (RFID)*, *Global Positioning System (GPS)*, and sensor technologies has triggered the development of the *Internet of Things (IoT)*. These new paradigms are only briefly introduced here. We will study the details of SOA in [Chapter 5](#); virtualization in [Chapter 3](#); cloud computing in [Chapters 4, 6, and 9](#); and the IoT along with cyber-physical systems (CPS) in [Chapter 9](#).

When the Internet was introduced in 1969, Leonard Klienrock of UCLA declared: “As of now, computer networks are still in their infancy, but as they grow up and become sophisticated, we will probably see the spread of computer utilities, which like present electric and telephone utilities, will service individual homes and offices across the country.” Many people have redefined the term “computer” since that time. In 1984, John Gage of Sun Microsystems created the slogan, “The network is the computer.” In 2008, David Patterson of UC Berkeley said, “The data center is the computer. There are dramatic differences between developing software for millions to use as a service versus distributing software to run on their PCs.” Recently, Rajkumar Buyya of Melbourne University simply said: “The cloud is the computer.”

This book covers clusters, MPPs, P2P networks, grids, clouds, web services, social networks, and the IoT. In fact, the differences among clusters, grids, P2P systems, and clouds may blur in the future. Some people view clouds as grids or clusters with modest changes through virtualization. Others feel the changes could be major, since clouds are anticipated to process huge data sets generated by the traditional Internet, social networks, and the future IoT. In subsequent chapters, the distinctions and dependencies among all distributed and cloud systems models will become clearer and more transparent.

### 1.1.1.5 Computing Paradigm Distinctions

The high-technology community has argued for many years about the precise definitions of centralized computing, parallel computing, distributed computing, and cloud computing. In general, *distributed computing* is the opposite of *centralized computing*. The field of *parallel computing*

overlaps with distributed computing to a great extent, and *cloud computing* overlaps with distributed, centralized, and parallel computing. The following list defines these terms more clearly; their architectural and operational differences are discussed further in subsequent chapters.

- **Centralized computing** This is a computing paradigm by which all computer resources are centralized in one physical system. All resources (processors, memory, and storage) are fully shared and tightly coupled within one integrated OS. Many data centers and supercomputers are *centralized systems*, but they are used in parallel, distributed, and cloud computing applications [18,26].
- **Parallel computing** In parallel computing, all processors are either tightly coupled with centralized shared memory or loosely coupled with distributed memory. Some authors refer to this discipline as *parallel processing* [15,27]. Interprocessor communication is accomplished through shared memory or via message passing. A computer system capable of parallel computing is commonly known as a *parallel computer* [28]. Programs running in a parallel computer are called *parallel programs*. The process of writing *parallel programs* is often referred to as *parallel programming* [32].
- **Distributed computing** This is a field of computer science/engineering that studies distributed systems. A *distributed system* [8,13,37,46] consists of multiple autonomous computers, each having its own private memory, communicating through a computer network. Information exchange in a distributed system is accomplished through *message passing*. A computer program that runs in a distributed system is known as a *distributed program*. The process of writing distributed programs is referred to as *distributed programming*.
- **Cloud computing** An *Internet cloud* of resources can be either a centralized or a distributed computing system. The cloud applies parallel or distributed computing, or both. Clouds can be built with physical or virtualized resources over large data centers that are centralized or distributed. Some authors consider cloud computing to be a form of *utility computing* or *service computing* [11,19].

As an alternative to the preceding terms, some in the high-tech community prefer the term *concurrent computing* or *concurrent programming*. These terms typically refer to the union of parallel computing and distributing computing, although biased practitioners may interpret them differently. *Ubiquitous computing* refers to computing with pervasive devices at any place and time using wired or wireless communication. The *Internet of Things* (IoT) is a networked connection of everyday objects including computers, sensors, humans, etc. The IoT is supported by Internet clouds to achieve ubiquitous computing with any object at any place and time. Finally, the term *Internet computing* is even broader and covers all computing paradigms over the Internet. This book covers all the aforementioned computing paradigms, placing more emphasis on distributed and cloud computing and their working systems, including the clusters, grids, P2P, and cloud systems.

#### 1.1.1.6 Distributed System Families

Since the mid-1990s, technologies for building P2P networks and *networks of clusters* have been consolidated into many national projects designed to establish wide area computing infrastructures, known as *computational grids* or *data grids*. Recently, we have witnessed a surge in interest in exploring Internet cloud resources for data-intensive applications. Internet clouds are the result of moving desktop computing to service-oriented computing using server clusters and huge databases

at data centers. This chapter introduces the basics of various parallel and distributed families. Grids and clouds are disparity systems that place great emphasis on resource sharing in hardware, software, and data sets.

Design theory, enabling technologies, and case studies of these massively distributed systems are also covered in this book. Massively distributed systems are intended to exploit a high degree of parallelism or concurrency among many machines. In October 2010, the highest performing cluster machine was built in China with 86016 CPU processor cores and 3,211,264 GPU cores in a Tianhe-1A system. The largest computational grid connects up to hundreds of server clusters. A typical P2P network may involve millions of client machines working simultaneously. Experimental cloud computing clusters have been built with thousands of processing nodes. We devote the material in [Chapters 4 through 6](#) to cloud computing. Case studies of HTC systems will be examined in [Chapters 4 and 9](#), including data centers, social networks, and virtualized cloud platforms

In the future, both HPC and HTC systems will demand multicore or many-core processors that can handle large numbers of computing threads per core. Both HPC and HTC systems emphasize parallelism and distributed computing. Future HPC and HTC systems must be able to satisfy this huge demand in computing power in terms of throughput, efficiency, scalability, and reliability. The system efficiency is decided by speed, programming, and energy factors (i.e., *throughput per watt* of energy consumed). Meeting these goals requires to yield the following design objectives:

- **Efficiency** measures the utilization rate of resources in an execution model by exploiting massive parallelism in HPC. For HTC, efficiency is more closely related to job throughput, data access, storage, and power efficiency.
- **Dependability** measures the reliability and self-management from the chip to the system and application levels. The purpose is to provide high-throughput service with Quality of Service (QoS) assurance, even under failure conditions.
- **Adaptation in the programming model** measures the ability to support billions of job requests over massive data sets and virtualized cloud resources under various workload and service models.
- **Flexibility in application deployment** measures the ability of distributed systems to run well in both HPC (science and engineering) and HTC (business) applications.

### 1.1.2 Scalable Computing Trends and New Paradigms

Several predictable trends in technology are known to drive computing applications. In fact, designers and programmers want to predict the technological capabilities of future systems. For instance, Jim Gray's paper, "Rules of Thumb in Data Engineering," is an excellent example of how technology affects applications and vice versa. In addition, Moore's law indicates that processor speed doubles every 18 months. Although Moore's law has been proven valid over the last 30 years, it is difficult to say whether it will continue to be true in the future.

Gilder's law indicates that network bandwidth has doubled each year in the past. Will that trend continue in the future? The tremendous price/performance ratio of commodity hardware was driven by the desktop, notebook, and tablet computing markets. This has also driven the adoption and use of commodity technologies in large-scale computing. We will discuss the future of these computing trends in more detail in subsequent chapters. For now, it's important to understand how distributed